

UNIT 1: INTRODUCTION (15 Q&A)

1. What is an Operating System? → It is system software that manages hardware and software resources.
 2. What are the main functions of an OS? → Process, memory, file, and device management.
 3. What are OS generations? → Batch, multiprogramming, time-sharing, modern OS.
 4. What is batch OS? → Executes jobs in batches without user interaction.
 5. What is time-sharing OS? → Allows multiple users to share CPU time.
 6. What is real-time OS? → Provides immediate response to inputs.
 7. What are types of OS? → Batch, time-sharing, distributed, real-time.
 8. What are OS services? → Program execution, I/O, file, communication.
 9. What is system call? → Interface between user program and OS.
 10. Types of system calls? → Process, file, device, information calls.
 11. What is layered OS? → OS divided into hierarchical layers.
 12. What is monolithic OS? → Entire OS works as a single unit.
 13. What is microkernel OS? → Minimal kernel with services in user space.
 14. What is virtual machine? → Software that simulates hardware system.
 15. Advantage of VM? → Isolation and resource sharing.
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UNIT 2: PROCESS, THREAD & SCHEDULING (20 Q&A)

16. What is a process? → Program in execution.
17. Process states? → New, ready, running, waiting, terminated.
18. What is state transition? → Movement between process states.
19. What is PCB? → Data structure storing process information.
20. Context switching? → Saving and loading process state.
21. Process relationship? → Parent-child relationship.
22. What is a thread? → Lightweight process.
23. Thread states? → Ready, running, blocked.
24. Benefits of threads? → Faster execution and resource sharing.
25. Types of threads? → User-level and kernel-level.
26. Multithreading? → Multiple threads in a process.
27. Process scheduling? → Selecting process for CPU execution.

- 28. Scheduling objectives? → Max CPU use, min waiting time.
 - 29. Types of schedulers? → Long, short, medium term.
 - 30. CPU utilization? → Percentage of CPU usage.
 - 31. Turnaround time? → Completion time – arrival time.
 - 32. Waiting time? → Time spent in ready queue.
 - 33. Response time? → First response delay time.
 - 34. FCFS? → First Come First Serve scheduling.
 - 35. SJF? → Shortest Job First scheduling.
 - 36. RR? → Round Robin with time quantum.
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UNIT 3: IPC (15 Q&A)

- 37. What is IPC? → Communication between processes.
 - 38. Critical section? → Shared resource access part.
 - 39. Race condition? → Multiple processes accessing data simultaneously.
 - 40. Mutual exclusion? → Only one process in critical section.
 - 41. Hardware solution? → Using atomic instructions.
 - 42. Strict alternation? → Processes take turns.
 - 43. Peterson's solution? → Software solution for two processes.
 - 44. Producer-consumer problem? → Synchronization of producer and consumer.
 - 45. Semaphore? → Synchronization variable.
 - 46. Types of semaphore? → Binary and counting.
 - 47. Event counter? → Tracks events for synchronization.
 - 48. Monitor? → High-level synchronization construct.
 - 49. Message passing? → Communication via messages.
 - 50. Reader-writer problem? → Managing read/write access.
 - 51. Dining philosopher problem? → Resource sharing issue.
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UNIT 4: DEADLOCK (10 Q&A)

- 52. Deadlock? → Processes waiting forever.
- 53. Necessary conditions? → Mutual exclusion, hold-wait, no preemption, circular wait.
- 54. Mutual exclusion? → Resource used by one process only.

- 55. Hold and wait? → Holding one resource while waiting.
 - 56. No preemption? → Resources cannot be taken forcibly.
 - 57. Circular wait? → Circular chain of processes.
 - 58. Prevention? → Eliminate one condition.
 - 59. Avoidance? → Safe resource allocation.
 - 60. Banker's algorithm? → Deadlock avoidance method.
 - 61. Detection & recovery? → Detect and terminate processes.
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UNIT 5: MEMORY MANAGEMENT (25 Q&A)

- 62. Memory management? → Managing main memory.
- 63. Logical address? → CPU-generated address.
- 64. Physical address? → Actual memory address.
- 65. Contiguous allocation? → Continuous memory allocation.
- 66. Fixed partition? → Equal memory blocks.
- 67. Variable partition? → Dynamic memory blocks.
- 68. Internal fragmentation? → Wasted space inside partition.
- 69. External fragmentation? → Free space scattered.
- 70. Compaction? → Combining free space.
- 71. Paging? → Dividing memory into pages.
- 72. Page frame? → Fixed block in memory.
- 73. Paging hardware? → Page table and MMU.
- 74. Protection? → Restrict unauthorized access.
- 75. Sharing? → Multiple processes share pages.
- 76. Disadvantage of paging? → Internal fragmentation.
- 77. Virtual memory? → Uses disk as memory extension.
- 78. Locality of reference? → Access nearby memory locations.
- 79. Page fault? → Page not in memory.
- 80. Working set? → Active pages set.
- 81. Dirty page? → Modified page.
- 82. Demand paging? → Load page when needed.
- 83. FIFO? → Replace oldest page.

- 84. Optimal? → Replace least used future page.
 - 85. LRU? → Replace least recently used page.
 - 86. NRU? → Replace not recently used page.
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UNIT 6: I/O, FILE & DISK (14 Q&A)

- 87. I/O device? → Hardware for input/output.
 - 88. Device controller? → Controls I/O devices.
 - 89. DMA? → Direct data transfer without CPU.
 - 90. Interrupt handler? → Handles interrupts.
 - 91. Device driver? → Software for hardware control.
 - 92. Device-independent I/O? → Common I/O interface.
 - 93. File? → Collection of data.
 - 94. Access methods? → Sequential, direct.
 - 95. File types? → Text, binary.
 - 96. File operations? → Read, write, delete.
 - 97. Directory structure? → File organization.
 - 98. Allocation methods? → Contiguous, linked, indexed.
 - 99. Free space management? → Bit vector, linked list.
 - 100. Disk scheduling? → FCFS, SSTF, SCAN, C-SCAN.
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